Strategic Intelligence Report: Orchestrating the Future of Precision Cell Therapy at the National Marrow Donor Program (NMDP)

Action-Oriented Executive summary

The National Marrow Donor Program (NMDP), following its comprehensive 2024 rebranding from Be The Match, is entering a transformative era where data science is the primary driver of its "Donor for All" initiative.[1] As the organization moves away from a binary model of Human Leukocyte Antigen (HLA) matching toward a multivariate predictive model of transplant success, the existing Business Intelligence (BI) infrastructure, primarily centered on Looker, faces significant white-space opportunities.[4] While Looker provides excellent structured visualization, it lacks the native capacity to process the 80-90% of healthcare data that remains unstructured—such as physician narratives, clinical trial protocols, and the global corpus of biomedical research.[6]

The most relevant and resonant use case for the NMDP BI and Analytics team is the development of a "Unified Transplant Outcome Intelligence & Predictive Research Engine".[3] This platform will concretely tie to Snowflake by leveraging the AI Data Cloud to marry structured CIBMTR outcomes with unstructured research data via Snowflake Cortex AI and Marketplace-native Cortex Knowledge Extensions (CKEs).[10] By integrating PubMed and ClinicalTrials.gov datasets, NMDP can build an agentic research assistant that provides cited, RAG-enabled insights directly within a Looker dashboard, solving for data latency and depth.[13]

The immediate next best action is the execution of a 1.5-hour hands-on lab focused on "Outcome Forecasting with Multimodal Data".[7] This lab will simulate a real-world request from the CIBMTR leadership to identify Graft-versus-Host Disease (GVHD) risk factors across diverse patient cohorts using a combination of internal registry data and external Social Determinants of Health (SDoH).[17] This intelligence request will transition the team from descriptive reporting to prescriptive analytics, aligning with the CIO's mandate to modernize and scale technology platforms.[20]

Company Snapshot

NMDP is a global nonprofit cell therapy leader managing a registry of 42 million+ potential donors to provide life-saving treatments for blood cancers and disorders.[1] Headquartered in Minneapolis with operations in Mexico, it facilitates thousands of transplants annually via a

network of 310+ research-participating centers.[2] A key 12-18 month shift is the "Donor for All" strategy, proving that 7/8 mismatched donors can achieve outcomes on par with 8/8 matches.[3]

Use Case Opportunity

Opportunity Name: Unified Transplant Outcome Intelligence & Predictive Research Engine

Why Now (Evidence)

NMDP's recent clinical trial data (ASH 2025) demonstrates a paradigm shift where mismatched unrelated donor (MMUD) transplants achieve survival rates >80%, fundamentally changing the standard of care.[3] The CIBMTR has expanded its "HCT Essentials Extract" to over 18,000 variables, creating a high-complexity data landscape that current BI tools cannot interpret in a narrative or predictive context without advanced AI orchestration.[26] Furthermore, search and procurement revenues have increased by 15%, demanding higher operational efficiency in donor-patient matching.[28]

Proposed Snowflake Play: App & AI on Governed Data

This play centers on using Snowflake Cortex AI as an orchestration layer to build a retrieval-augmented generation (RAG) system.[8] It leverages Snowflake Cortex Search to index unstructured clinical notes and the PubMed Cortex Knowledge Extension for external research.[10] The solution will be visualized in Looker via a "Connected App" architecture, allowing users to interact with high-fidelity analytics and AI-generated insights through a unified semantic layer.[4]

Required Data & Owners

Data Source	System/Type	Likely Owner/Team
Registry & Donor Data	NMDP Registry (Structured)	Jamie Margolis (SVP, Donor & Product Ops) [31]
Clinical Outcomes	CIBMTR Outcomes Database (18k+ Vars)	Steven Devine (CMO & CIBMTR Exec Lead) [31]
Unstructured Notes	Physician Narratives/Interaction Logs	Julie Smolich (SVP, Patient & Provider Services) [31]

Research Corpus	PubMed CKE (Snowflake Marketplace)	BI & Analytics Team [10]
SDoH Data	AnalyticsIQ / Equifax (Marketplace)	Tracy Smith (Sr. Director, Data & Analytics) [18]

The Strategic Evolution of NMDP: From Registry to Research Powerhouse

The National Marrow Donor Program has undergone a profound strategic shift over the past 36 months, moving from its legacy identity as "Be The Match" to the unified, modernized "NMDP" brand.[1] This rebranding is not merely cosmetic; it signals an organizational pivot toward a comprehensive cell therapy ecosystem that encompasses biotherapies, cryopreservation services, and advanced implementation science.[2] The historical mission, established in 1987 with the success of the first unrelated donor transplant, was centered on the 8/8 HLA match—a "needle in a haystack" search that often left diverse patients without options.[2] Today, the organization's "Donor for All" initiative, supported by findings from the Phase II ACCESS trial, has proven that nearly 100% of patients with common blood cancers can now find a suitable donor through the use of mismatched unrelated donor (MMUD) grafts paired with post-transplant cyclophosphamide (PTCy).[3]

This evolution places unprecedented pressure on the Business Intelligence and Analytics team. The variables that dictate transplant success are no longer just genetic; they are now operational, financial, and social.[7] The CIBMTR (Center for International Blood and Marrow Transplant Research), a collaboration between NMDP and the Medical College of Wisconsin, currently manages a research portfolio of over 185 ongoing clinical trials and has recently expanded its "HCT Essentials Extract" to include more than 18,000 collected and derived variables.[19] Analyzing this volume of data to find "genotype-phenotype correlations" or to predict GVHD risk requires a move away from static dashboarding toward a dynamic, AI-enabled data architecture.[9]

Operational Context and the "Looker Gap"

The current BI landscape at NMDP leverages Looker as its primary visualization interface, often paired with Snowflake as the underlying data warehouse and dbt for transformations.[33] Looker's strength lies in its "LookML" semantic modeling layer, which provides a single source of truth for business metrics and allows for "Google-easy" dashboarding.[4] However, the NMDP BI team faces a significant challenge: "Looker's native limitations in handling unstructured and semi-structured biological data".[4]

Clinical documentation, which makes up roughly 80% of healthcare data, exists in physician

notes, discharge summaries, and clinical narratives.[7] Looker is essentially a SQL generator; it cannot natively "read" these unstructured formats or perform complex natural language processing (NLP) to extract sentiment or clinical markers without pre-processing.[5] This is where Snowflake creates a superior set of insights. By using Snowflake Cortex AI, NMDP can process these unstructured narratives directly in the warehouse, transforming them into structured "tokens" or "embeddings" that Looker can then visualize as part of a broader dashboard.[8]

Comparison of Modern Data Stack Capabilities for NMDP

Feature	Looker (Current State)	Snowflake + Cortex (Proposed State)
Structured Data Visualization	Exceptional via LookML [4]	High-performance via SQL/Snowsight [35]
Unstructured Narrative Processing	Limited; requires prior ETL/flattening [5]	Native via Cortex AI (Summarize/Extract) [9]
External Research Integration	API-based; often latent [4]	Native via Cortex Knowledge Extensions [10]
Predictive Modeling (ML)	Limited to simple trend lines [4]	Advanced via Snowpark & Snowflake Notebooks [7]
Semantic Search	No native vector search	Native via Cortex Search [13]

Financial Stewardship and Revenue Streams

For the fiscal year ending September 30, 2024, NMDP reported a total revenue of approximately $667 million, a substantial increase from the $578 million reported in 2023.[28] This growth is primarily fueled by Search and Procurement fees, which rose from $486 million to over $560 million in a single year.[28] This 15% growth underscores the increasing volume of cell therapy interventions and the organizational need for scalable data solutions to manage the logistics of donor collection and product delivery.[22]

The organization's expenses also grew, totaling $623 million in 2024.[28] A significant portion of these funds is dedicated to Patient Assistance, with NMDP providing over $6.6 million in grants to 3,300+ patient families to overcome financial barriers like housing and transportation.[22] For the BI team, this presents a "Cost & FinOps control" opportunity: using predictive analytics to identify patients at high risk of "financial toxicity" or "lost to follow-up" based on social

determinants of health (SDoH), thereby optimizing the allocation of donor-funded grants.[16]

Condensed Statement of Activities (FY2024 vs. FY2023)

Values in USD Thousands ($000s).[28]

Revenue Category	2024 Actual	2023 Actual	% Change
Search & Procurement Fees	$560,350	$486,414	+15.2%
Federal Contracts & Grants	$50,341	$55,427	-9.2%
Contributions/Philanthropy	$35,487	$35,274	+0.6%
Other Income	$20,490	$870	+2,255%
Total Revenue	**$666,668**	**$577,985**	**+15.3%**

The "Other Income" surge is likely related to investments and subsidiary activities like NMDP BioTherapies, which extends services to organizations developing new cell treatments.[1] The stability of search and procurement revenue makes NMDP a resilient "account" that is now investing heavily in technical modernizations.[20]

Use Case Deep Dive: The Predictive Research Engine

The most impactful showcase for the NMDP BI team is a dashboard that doesn't just show *what* happened (descriptive), but explains *why* it happened by querying external research (agentic).[13] Traditional dashboards at NMDP might show survival rates by donor type. The "top-notch" dashboard proposed here would use Snowflake Cortex to analyze physician notes from post-transplant follow-ups and automatically correlate them with the latest findings from the *PubMed Biomedical Research Corpus*.[9]

Technical Mechanism: Cortex Knowledge Extensions (CKEs)

Snowflake recently released Cortex Knowledge Extensions, which are essentially "Cortex Search" services shared via the Marketplace.[12] For NMDP, the PubMed and ClinicalTrials.gov CKEs allow the BI team to build a Retrieval-Augmented Generation (RAG) system without the

heavy lift of managing vector databases or complex ETL.[10]

A researcher using the Looker dashboard could click on a "high-risk" patient segment. The Snowflake backend would then:

1. Extract the specific clinical markers (e.g., HLA-DPB1 mismatch) from the patient's record.[9]
2. Pass these markers as a query to the OA_COMM.PUBMED_OA_CKE_SEARCH_SERVICE.[10]
3. Return a summarized paragraph explaining the latest research on that specific mismatch's impact on GVHD, cited with links to the original articles.[10]

This "Customer 360" for patients—integrating registry data, clinical outcomes, and the global research corpus—is something Looker cannot do alone, making Snowflake the indispensable engine of the intelligence pack.[5]

Snowflake Marketplace: Enhancing the Intelligence Agent

To build the very best hands-on lab or showcase, the BI team must leverage Marketplace assets that augment internal data with socioeconomic and biomedical context.[39]

Prioritized Marketplace Inventory for NMDP

Data Product	Provider	Value Proposition
PubMed Biomedical Research Corpus	Snowflake	Direct access to 4M+ research articles for RAG-enabled clinical agents.[10]
Clinical Trials Research Database	Snowflake	Real-time tracking of global studies to align NMDP BioTherapies with trial demand.[11]
Social Determinants of Health (SDoH)	AnalyticsIQ	Insights into economic stability and education to predict patient "financial toxicity".[18]
Healthcare SDoH Insights	Equifax	Postcode-level morbidity and causality insights to

		map "care deserts".[40]
AACT Clinical Trials Data	Element Data	Structured metadata for longitudinal analysis of intervention effectiveness.[43]
SVI (Social Vulnerability Index)	Sparksoft	Maps environmental burdens and health vulnerability to improve equitable recruitment.[44]

The use of these datasets allows the BI team to "build an agent in and on Snowflake" that acts as a subject matter expert for the CIBMTR researchers.[8] For example, the *PubMed CKE* provides a "search_preview" function that can be used directly in SQL to ask questions like: *"What lifestyle factors most strongly influence disease progression in multiple sclerosis?"*.[10]

High-Value Outreach and Organizational Hierarchy

Engagement with NMDP should focus on the intersection of "platform modernization" and "clinical leadership." The organization recently hired new C-suite leaders specifically to drive their five-year strategic plan.[20]

Prioritized Contact List

Name	Role	Rationale for Contact	Source Link
Jim Graham	Chief Information Officer	Joined July 2024; task is to modernize and scale platforms. Former CIO at Prime Therapeutics. Responsible for "speed-to-market" and automation.	20
Tracy Smith	Sr. Director, Data & Analytics	Lead for the "modern data stack" initiative. Worked with DAS42	33

		to architect enterprise solutions on Snowflake.	
Steven Devine, MD	Chief Medical Officer	Executive Lead for CIBMTR. The primary "customer" of the BI team's clinical research outputs. Focuses on MMUD survival.	3
Katherine Heyerdahl	Chief Financial Officer	Focuses on financial sustainability and pricing for products/services. Interested in ROI of data modernization.	20
Jamie Margolis, PhD	SVP, Donor & Product Ops	Oversees registry and collection. Interested in supply chain optimization for "just-in-time" cell therapy.	31
Erica Jensen	SVP, Strategy & Advancement	Drives the 5-year plan and awareness campaigns. Interested in using data for "equitable access" stories.	20
Julie Smolich	SVP, Patient & Provider Services	Owns the patient journey and support programs. Likely owner of the "unstructured" interaction data.	31

Sales Signal: The CIO's Mandate

Jim Graham's background at Prime Therapeutics involved a multiyear initiative to modernize and scale technology platforms to improve access to information.[20] His current role at NMDP explicitly includes "modernizing and scaling platforms and maintaining system security, automation and operational efficiencies".[20] This is a prime signal that NMDP is moving away from fragmented systems toward a unified AI Data Cloud.[7]

Blueprint for a 1.5-Hour Hands-On Lab

To resonate with the NMDP BI and Analytics team, the lab must feel like a "real-world" intelligence request from the CIBMTR.[19] The lab should demonstrate the transition from structured dashboarding to AI-augmented research.[8]

Lab Title: Orchestrating the "Donor for All" Intelligence Agent

0:00 - 0:15: The Modern Data Stack Setup

- Introduction to the Snowflake-Looker-dbt architecture.[35]
- Connecting Looker to Snowflake and defining the "LookML" semantic layer for the CIBMTR *HCT Essentials Extract*.[5]

0:15 - 0:45: Integrating the Marketplace & Unstructured Data

- "One-click" acquisition of the *PubMed Biomedical Research Corpus* CKE from the Snowflake Marketplace.[10]
- Task: Use Snowflake Cortex EXTRACT_ANSWER on a batch of simulated "Physician Notes" to identify secondary outcomes like "Cytokine Release Syndrome" (CRS).[9]
- Marrying these AI-extracted markers with structured patient demographics in a Snowflake table.[7]

0:45 - 1:15: Agentic RAG Workflows

- Building a "Cortex Agent" in Snowflake that allows researchers to ask: *"What are the emerging drug targets for treating GVHD in 7/8 matched patients?"*.[8]
- Using the snowflake.cortex.search_preview function to pull cited abstracts from PubMed directly into a Snowflake Notebook.[10]

1:15 - 1:30: The Superior Looker Experience

- Visualizing the combined data in Looker.[4]
- Final Output: A dashboard that displays survival forecasts (Predictive) alongside a "Research Chat" box (Agentic) that cites PubMed to explain the trends.[13]
- Demonstrating how Looker's "Conversational Analytics" can now answer deep clinical questions because Snowflake is providing the unstructured context.[4]

Conclusion: The Path to Predictive Maturity

NMDP is at the precipice of "unparalleled growth".[20] The move to "Donor for All" is a massive technical hurdle that cannot be solved by traditional BI alone.[3] By leveraging Snowflake as the orchestration layer for AI, Marketplace data, and unstructured clinical narratives, the NMDP BI and Analytics team can transform itself from a cost center into a "Clinical Innovation Hub".[7] This strategy not only satisfies the CIO's modernization mandate but directly accelerates the organization's mission to find life-saving cures for every patient, regardless of their background or genetic complexity.[1]

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